TOPIC 2 Transforming and Comparing Linear Functions

Name _____ Date _____

Reminders

Slope = $(y_2 - y_1) \div (x_2 - x_1)$ · $y = mx + b$ · point-slope: $y - y_1 = m(x - x_1)$

Parallel: same slope · Perpendicular: flip the fraction and change the sign (product −1)

Outside f(x) + k: up/down · Inside f(x − h): right h, f(x + h): left h (opposite the sign)

Horizontal: y = #, slope 0 · Vertical: x = #, slope undefined (HOY VUX)

Domain = inputs (x) · Range = outputs (y) · Zero = x-intercept

1. The graph of linear function m passes through (−4, 6) and (2, −3). Given $f(x) = x$ and $m(x) = af(x)$, what is the value of a?

Tip: a is the slope. Use $(y_2 - y_1) \div (x_2 - x_1)$.

Slope = (−3 − 6) ÷ (2 − (−4)) = ____ ÷ ____

a = _____

***a* = −3/2**

a is the slope: (−3 − 6) ÷ (2 − (−4)) = −9/6 = −3/2.

2. A food truck rents for $120 plus $45 per hour, for at most 6 hours. What is the range for one rental?

> **Tip:** Range = the y-values (dollars). Find the cost at 0 hours and at 6 hours.

F. $120 \le y \le 390$

G. $0 \le x \le 6$

H. $45 \le y \le 390$

Range = costs. 0 hours: $120. 6 hours: 120 + 45(6) = $390.

3. A vertical line passes through (8, −3). What are its equation and slope?

> **Tip:** HOY VUX: Vertical → Undefined slope → x = #.

A. $y = -3$; slope is zero

B. $x = 8$; slope is undefined

C. $x = 8$; slope is zero

Vertical: every point has x = 8. Slope = rise ÷ 0: undefined (VUX).

4. Lena graphed $f(x)$ and $g(x) = f(x + 4)$. Which statement describes how the graphs are related?

> **Tip:** Inside the parentheses: move the opposite way of the sign.

F. translated right 4 units

G. translated up 4 units

H. translated left 4 units

Inside the parentheses the graph moves opposite the sign: + 4 → left 4.

5. $f(x)$ has the table (−1, 8), (0, 5), (1, 2), (2, −1). The graph is translated 3 units left. What is the y-intercept of the new function?

> **Tip:** Left 3: subtract 3 from each x. Which point lands on x = 0?

Left 3: g(x) = f(x + 3)

g(0) = f(____) = _____

y-intercept = _____

−4

f(x) = −3x + 5. Left 3: g(x) = f(x + 3) = −3x − 4. Check: the point at x = 3, (3, −4), moves to (0, −4).

6. $f(x) = x$ and $g(x) = f(\frac{1}{2}x)$. Write the equation for $g(x)$.

> **Tip:** f(½x): replace x with ½x.

f(x) = x, so f(½x) = _____

g(x) = _____

g(x) = ½x

Put ½x in for x: g(x) = ½x, a less steep line through (0, 0).

7. Write the equation in slope-intercept form of the line that crosses the x-axis at 8 and is perpendicular to $y = 4x + 1$.

> **Tip:** Flip the slope and change the sign. Crosses the x-axis at 8 = the point (8, 0).

Slope = _____

0 = ____(8) + b → b = _____

y = _____x + _____

y = −1/4x + 2

Slope −1/4. Point (8, 0): 0 = −1/4(8) + b → b = 2.

8. Write the equation in slope-intercept form of the line through (−6, 0) that is parallel to $y = 1.5x - 4$.

> **Tip:** Parallel → same slope. Then put in the point and solve for b.

Slope = _____

0 = 1.5(−6) + b → b = _____

y = _____x + _____

***y* = 1.5*x* + 9**

Same slope 1.5. 0 = 1.5(−6) + b → b = 9.

9. A line passes through (1, −2) and (4, 7). Write an equation of the line in point-slope form.

> **Tip:** Find the slope, then use $y - y_1 = m(x - x_1)$.

Slope = (7 − (−2)) ÷ (4 − 1) = _____

y − 7 = ____(x − 4)

***y* − 7 = 3(*x* − 4) (*or y* + 2 = 3(*x* − 1))**

Slope = (7 − (−2)) ÷ (4 − 1) = 9 ÷ 3 = 3. With (4, 7): y − 7 = 3(x − 4).

10. A line passes through (0, 6) and (3, 0). Which statement is true?

> **Tip:** Zero = x-intercept. Decreasing = goes down left to right.

F. It is decreasing and its zero is 6.

G. It is increasing and its zero is 3.

H. It is decreasing and its zero is 3.

It falls left to right (slope −6/3 = −2). It crosses the x-axis at 3. The y-intercept is 6.

11. Which pair of equations represents perpendicular lines?

> **Tip:** Multiply the slopes: −1 means perpendicular.

A. $y = 2/3x - 1$ and $y = -3/2x + 4$

B. $y = 2/3x + 1$ and $y = 2/3x - 4$

C. $y = 2/3x$ and $y = 3/2x + 2$

$2/3 \times (-3/2) = -1$. (The second pair is parallel.)

12. A phone is at 80% battery and loses 4% per hour until it is empty. What is the domain?

> **Tip:** Domain = x-values (hours). How long until 0%?

F. $0 \leq x \leq 80$

G. $\mathbf{0 \leq x \leq 20}$

H. $0 \leq y \leq 80$

Domain = hours. 80 ÷ 4 = 20 hours.